where P_k is the k^{th} rule of fuzzification, and x_j and Bk_j are fuzzy features and fuzzy sets. The membership functions of fuzzy of Bk_j, normally distributed, are as follows:

$$\mu_{ki}(n_i) = \exp\left(-\left(\frac{n_i - d_{ki}}{\sigma_{ki}}\right)^2\right) \tag{6.8}$$

Additionally, d_{ki} illustrates the central points, and σ_{ki} shows the standard deviation of membership function related to k. The results of the TSK fuzzy method with M rules are as follows:

$$m = \frac{\sum_{k=1}^{y} m_k \prod_{i=1}^{x} \mu_{ki}(n_i)}{\sum_{k=1}^{y} \prod_{i=1}^{x} \mu_{ki}(n_i)} \tag{6.9}$$

6.3.5.3 Wavelet Transformation

In signal processing, wavelet algorithms are a crucial tool. Large volumes of data may be analyzed for patterns with this technique. Using time series and neural networks, it is necessary to do modeling activities to address the prediction issue. Non-linearity may be difficult to forecast using neural networks as a general predictor.

In the frequency domain, wavelets may simultaneously depict functions and expose their local properties. Neural networks may be trained to properly predict extremely non-linear data because of these properties. Here is a definition of a wavelet:

$$\psi_{c,d} = |c|^{-\frac{1}{2}} \psi\left(\frac{n-d}{c}\right), (c, d \epsilon P, c \neq 0) \tag{6.10}$$

Equatorial wavelet generator derived from this equation:

$$D_\psi = \int_0^{+\infty} \frac{|\hat{\psi}(u)|^2}{u} du < +\infty \tag{6.11}$$

To mimic multivariable processes, wavelets must be defined in many dimensions. This chapter defines multidimensional wavelets in the following manner:

$$\psi_j(n) = \prod_{i=1}^{x} \psi\left(\frac{n_i - d_{ji}}{c_{ji}}\right),\ j = 1, 2, \ldots, X, \tag{6.12}$$

$d_{j=d_{ji}}$ is translation and $c_j = (c_{ji})$ is dilation

$$m_y = \sum_{j=1}^{x} u_{yj} \Psi_j + \bar{m}_y, y = 1, 2, \ldots, v, \tag{6.13}$$

The following equation combines wavelet and neural network components.

6.3.5.4 Wavelet Fuzzy Logic-Based Deep Neural Network

The overall structure of the WFL-DNN rules was ambiguous.

$$\begin{gathered} \text{IF } n_1 \text{ is } B_{k1....} \text{ AND } n_x \text{ is } B_{kx} \\ \text{THEN} \mathrm{M}_k = \sum_{j=1}^{x_k} u_j^k \, \psi_j^k + \bar{\mathrm{M}}_k \end{gathered} \tag{6.14}$$

In the last portions of the rules, many WNNs with Nk wavelet activation functions were utilized. First, layer 1 was the input feature, and layer 2 was the first hidden layer of the combination that was shown. Fuzzy membership function produced the second layer, according to equation (6.15).

$$\Psi_j^k = \prod_{i=1}^{x} \psi_{ji}^k = \prod_{i=1}^{x} \psi\left(\frac{n_i - d_{ji}^k}{C_{ji}^k}\right) \tag{6.15}$$

There was a fuzzy rule for every node in layer 3. The AND operator was used to compute the attributes that were gleaned from the data.

$$m_k = O_k^{(4)} \sum_{j=1}^{X_k} u_j^k \Psi_j^k + \bar{m}_k \tag{6.16}$$

Defuzzification conclusions were covered in depth in Layers 5 and 7. Layer 4 doubled the Layer 3 output data. As a result, the output signals from Layers 5 and 3 were summarized by two neurons in Layer 6. Layer 7's output neuron created the quotient, which represented the ratio of the output of each WNN to that of the final WFL-DNN as shown in Figure 6.7.

$$O_k^{(5)} = O_k^{(3)} . O_k^{(4)} = O_K . m_k \tag{6.17}$$

$$O_1^{(6)} = \sum_{K=1}^{Y} O_k^{(5)}, O_2^{(6)} = \sum_{K=1}^{y} O_k^{(3)} \tag{6.18}$$

$$y = O^{(7)} = \frac{O_1^{(6)}}{O_2^{(6)}} = \frac{\sum_{K=1}^{y} O_K m_k}{\sum_{K=1}^{y} O_K} \tag{6.19}$$

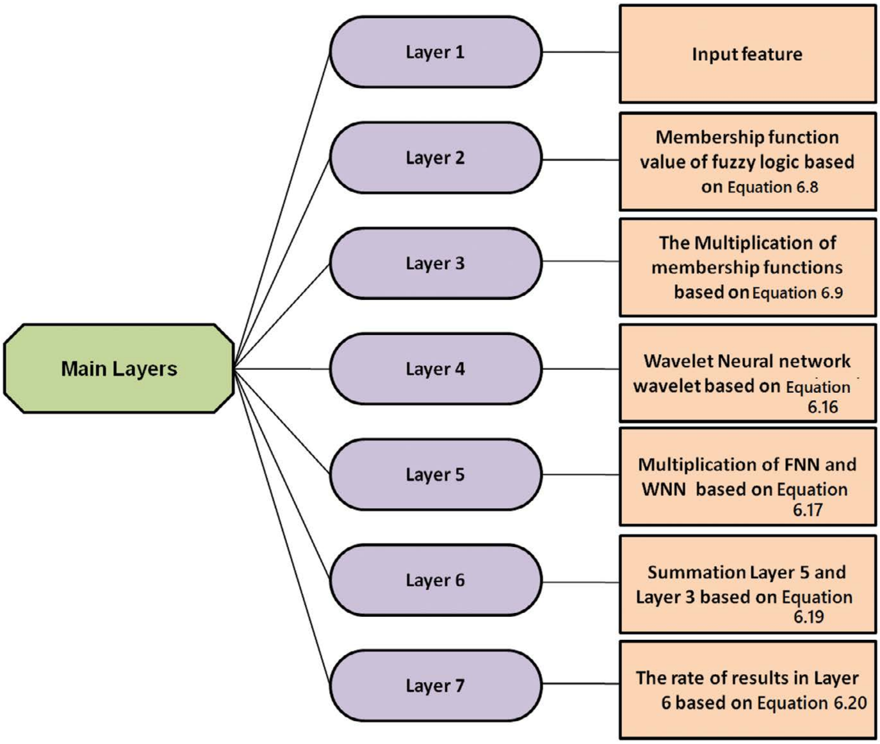

FIGURE 6.7 The WFL-DNN method's architecture.

Variable calibration of the WFL-DNN led to the usage of gradient descent to train it. The objective variable's gradient was flipped around such that the set points were:

$$E(\Theta, n, m) = \frac{1}{2}(m - f)^2, \tag{6.20}$$

$$\Delta\Theta = \left(\begin{array}{c} \Theta(t+1) = \Theta(t) + \Delta\Theta \\ -\gamma d \dfrac{\partial E}{\partial d_{ki}}, -\gamma\sigma \dfrac{\partial E}{\partial d_{ki}}, -\gamma a \dfrac{\partial E}{\partial a_{ji}^{k}}, -\gamma a \dfrac{\partial E}{\partial b_{ji}^{k}}, \\ -\gamma u \dfrac{\partial E}{\partial u_{j}^{k}}, -\gamma \bar{m} \dfrac{\partial E}{\partial \bar{m}_{k}} \end{array} \right) \tag{6.21}$$

6.4 PERFORMANCE ANALYTICS

Data collection by organizations has outgrown the capabilities of conventional HRM systems, which are no longer able to handle data and make choices. Using data mining techniques, HRM can benefit from discovering new information (Khang, Rani, & Shah, 2024).

WFL-DNN was used in this study to establish a framework for HR managers to better track employee performance and make more informed choices about HRM strategies. In this study, the parameters used are accuracy, precision, recall, decision-making skill, and execution time.

The findings were compared to those obtained using existing approaches. The existing methods, such as artificial neural networks (ANNs), long short-term memory-convolutional neural networks (LSTM-CNNs), and back propagation neural networks (BP-NNs) are compared with the proposed method to attain the greatest performance in this work.

6.4.1 Accuracy

In the testing phase, accuracy measures how many accurate predictions a classifier produced concerning the label's actual value. Also known as the number of accurate assessments to the total number of assessments, it may be expressed as a ratio. Equation (6.22) is used to determine accuracy (Figure 6.8).

$$\text{Accuracy} = \frac{(\text{TN} + \text{TP})}{(\text{TN} + \text{TP} + \text{FN} + \text{FP})} \tag{6.22}$$

where

- TN = True Negative.
- TP = True Positive.
- FP = False Positive.
- FN = False Negative.

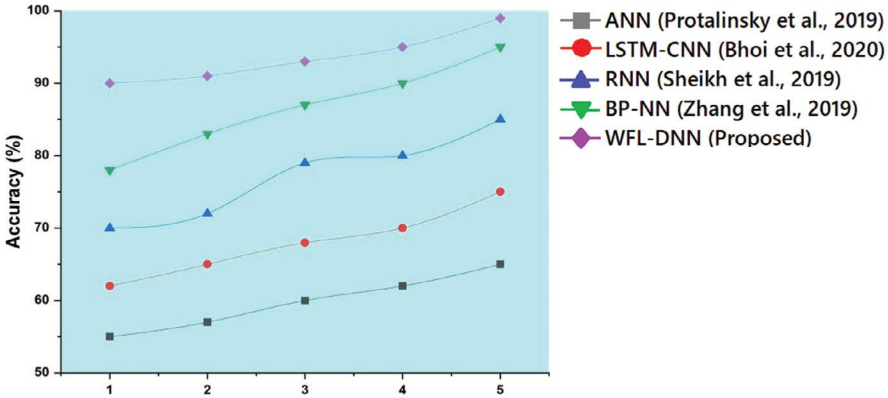

FIGURE 6.8 Comparison of accuracy for existing and proposed methodologies (Bhoi et al., 2020; Protalinsky et al., 2019; Sheikh et al., 2019; Zhang et al., 2019).

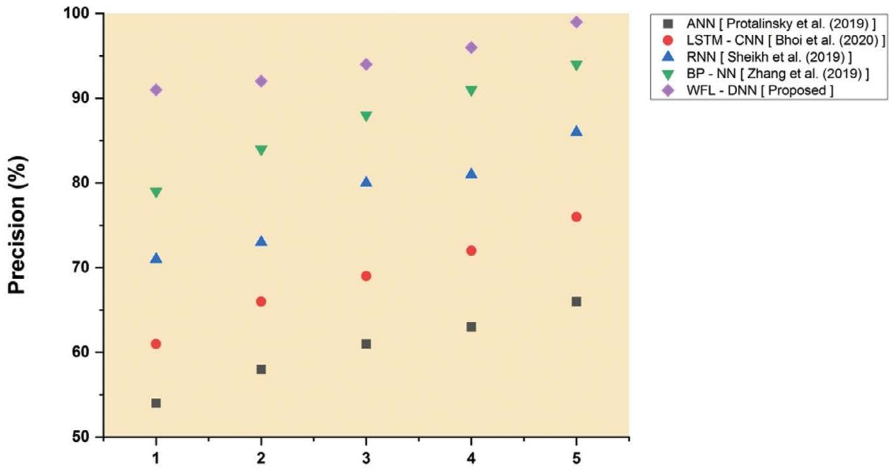

FIGURE 6.9 Comparison of precision for existing and proposed methodologies (Bhoi et al., 2020; Protalinsky et al., 2019; Sheikh et al., 2019; Zhang et al., 2019).

6.4.2 Precision

Precision is a key metric for assessing accuracy; it expresses the proportion of cases that the classifier classified as positive relative to the total instances of predictively positive data as indicated in equation (6.23) (Figure 6.9).

$$\text{“Precision} = \frac{TP}{TP + FP}\text{”} \tag{6.23}$$

6.4.3 Recall

Recall establishes completeness of the proportion of occurrences that the classifier correctly classified as positive. When there is a large cost associated with false negative, the recall is a performance parameter used to choose the optimal model (Figure 6.10).

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6.24}$$

6.4.4 Decision-Making Skill

The capacity to make decisions refers to the set of abilities that help in the process of selecting appropriate responses to problems. With these abilities, you will be able to make educated judgments after gathering all of the pertinent facts and data and taking into consideration a variety of perspectives (Figure 6.11).

6.4.5 Execution Time

System services and run-time processes are included in the definition of a specific task's execution time, which is defined as the time spent by the system performing

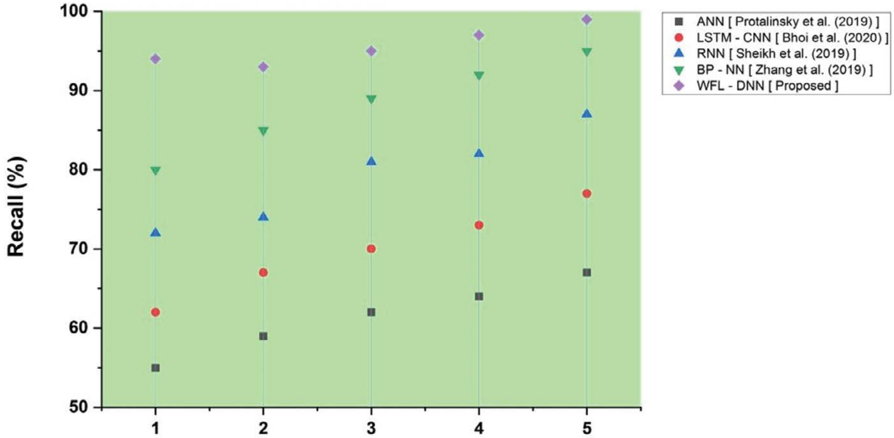

FIGURE 6.10 Comparison of recall for existing and proposed methodologies (Bhoi et al., 2020; Protalinsky et al., 2019; Sheikh et al., 2019; Zhang et al., 2019).

that task. Implementation-specific time-tracking mechanisms are employed to measure execution time (Figure 6.12).

The proposed method WFL-DNN has 58 seconds. The findings of the studies indicate that the suggested approach requires a shorter amount of time than that which is needed by the job to finish its execution (Khang et al., 2024).

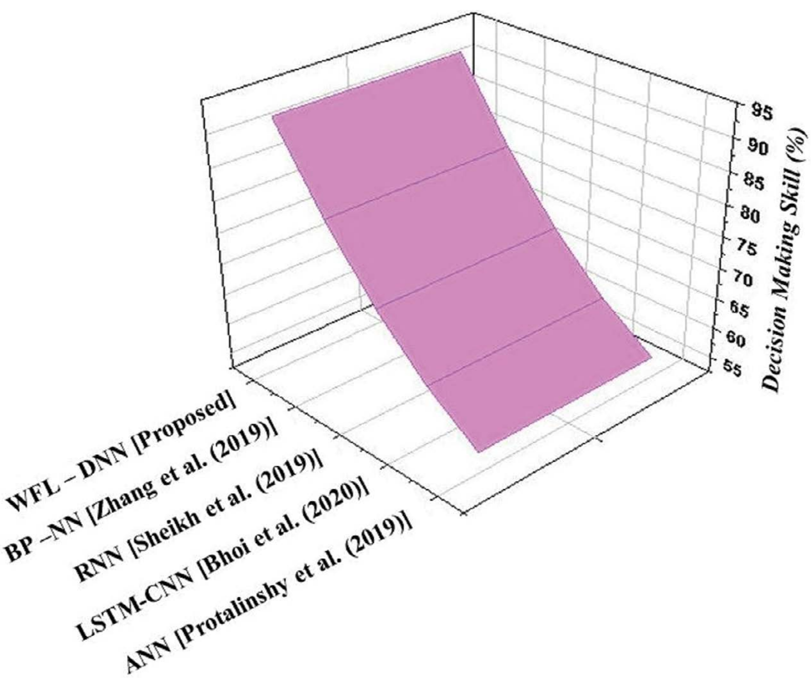

FIGURE 6.11 Comparison of decision-making skills for existing and proposed methodologies.

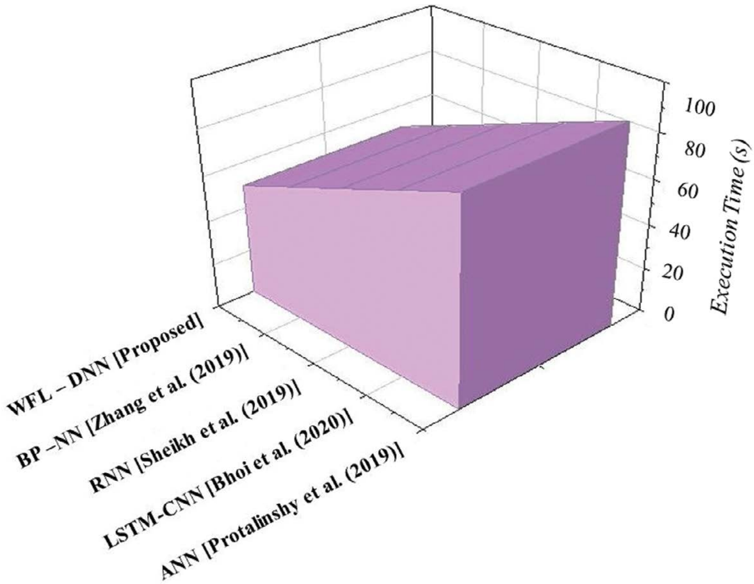

FIGURE 6.12 Comparison of execution time for existing and proposed methodologies.

6.5 CONCLUSION

HRM initiatives constitute the foundation of the information system. HRM information system design and implementation are described in general terms.

Management information systems are critical to the development of a human resources information system, which is based on personnel management, to guarantee flexibility in using methods and effective management of the internal market.

Personnel management is the process of organizing, coordinating, and monitoring operations, as well as the acquisition, development, and remuneration of human resources, to fulfill a variety of personnel and organizational objectives in today's globalized world. The term "personnel" refers to the individuals who work for a company or organization to accomplish a certain purpose (Khang et al., 2022a).

Results show that the DNN strategy is effective in terms of accuracy with 99%, precision with 99%, recall with 99%, decision-making skill with 92%, and execution time with 58 seconds, and that it beats the classic ANN, LSTM-CNN, RNN, and BP-NN (Khang et al., 2022b).

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7 Data-Driven Application of Human Capital Management Databases, Big Data, and Data Mining

Alex Khang, Shashi Kant Gupta, Chandra Kumar Dixit, and Parin Somani

CONTENTS

7.1 INTRODUCTION

Today's business market competitiveness is getting stronger and more globalized (Zeng, 2021). The majority of business owners are aware that of all the resources available to them, human resources are the most dynamic and demanding.

The business creates a human resource management (HRM) organization made up of corporate managers, and the supervisor and groups of stakeholders in charge

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of managing human resources put the idea of dynamism into practice in a systematic and all-encompassing way (Meena & Girija, 2022).

An efficient tool is required to aid managers in administration as the demand for human resource managers and the level of specialization of human resource departments both keep rising. Data mining technology, which can extract underlying principles from recorded data and use them as a foundation for judgment, is also advancing (Dayananda & Silva, 2020).

The application of critical corporate international sustainability is cited as the main motivator for improving internal and technological sustainability (Islami & Mulolli, 2021). Organizations tend to encourage their staff to think analytically to use the resources of the company and maintain them for financial success.

In the current environment, innovation is seen as the primary component for development and prosperity in the entity (Li, 2021). By integrating corporate practices into the HRM process, such as employee recruitment and selection, learning, and work system implementation, HRM helps to achieve sustainable development.

Employers and experts can show a helpful connection between increasing corporate sustainability and teamwork in overall environmental management initiatives (Gui & Zhang, 2022). Organizations now need to establish ecological policies and programs due to growing environmental awareness and the acceptance of international environmental protection standards.

HRM provides ecologically responsible activities, including internet recruitment, safeguarding digital records, conducting extensive interviews and supporting documents, receiving feedback and coaching, and receiving incentives based on initiatives (Wang, 2021).

The primary goal is to determine if the area of HRM can increase sustainable development by implementing the HRM concept in organizations. Additionally, this survey aids in examining the HRM practices already in use within the business, as well as perceptions of such practices and related barriers as reported by the organizational experts (Zhang, 2021).

The straight answer to how an organization can accomplish high results, how it can ensure a sustainable strategic advantage, how it can compete in the marketplace, and how it can improve organizational effectiveness for a long period of time are all questions related to human resources as a valuable asset of an organization (Zhang et al., 2021).

An organization's primary competitive advantage originates from the value materials it possesses, which may be physical but frequently are intellectual capital, such as reputation, talents, brands, or something similar, which are difficult for rivals to replicate (Xu, Tu & Xiao, 2022). Big data technology has made a significant impact on daily lifestyles and may be used methodically in businesses to contribute significantly to the original (Liu, 2021) as shown in Figure 7.1.

Corporate human resource managers should have a big data mindset, have exceptional talent distribution skills, have forward-thinking awareness when forming strategies, commit to everyday managerial activities, and put design abilities into effect. As a result, HR work must rely increasingly on cutting-edge technological platforms to collect data, enhance analytical abilities, and do more advising, monitoring, and judgment responsibilities (Khang et al., 2023).

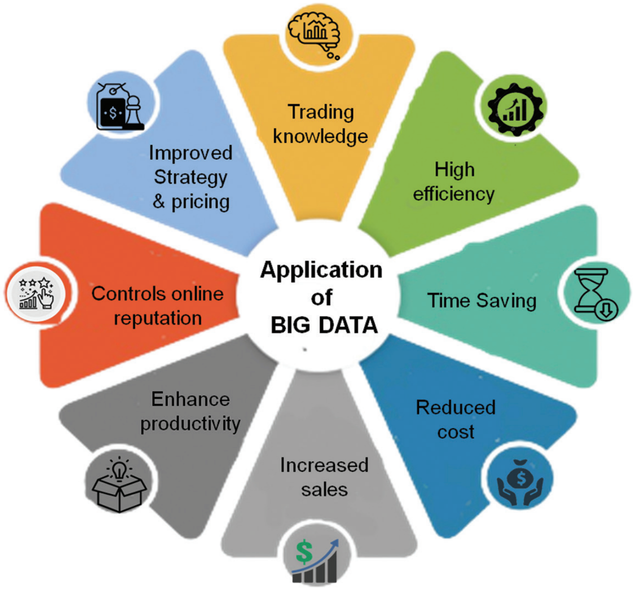

FIGURE 7.1 Application of big data in HRM.

Big data applications span a variety of academic fields, including analytics, computer programming, and economics (Zhang, 2021). They also call for a change in the skills required for HR roles. The research shows that practitioners of HRM need to develop their diverse technologies, better productivity, and other adaptability. Nowadays, the emergence of online digital technologies further encourages the use of big data in a variety of businesses (Babasaheb, Sphurti & Khang, 2023a). Big data and online information technology coexist (Gupta et al., 2023).

The conventional human resources performance approach to management has altered in the setting of big data (Wei & Jin, 2021). Analysis of the data may be used to monitor human resource productivity from the perspective of big data, as opposed to depending exclusively on the executives' perception. Achievement management has the power to ignite employee passion and boost an organization's bottom line. The execution of traditional performance management, however, is fraught with issues.

Optimizing the method of managing the effectiveness of human resources is thus the primary defensive measure in the context of big data to follow societal trends and encourage the positive growth of the business.

The arrival of 5G communications systems has changed the media sector's trajectory of growth (Zhao & Xue, 2021). This means that to satisfy the needs of 5G communications system distribution in the age of big data, 5G communication networks need to be continually modified. This examines how big data is used in the 5G communication network.

The goal is to prevent data anomalies from happening and to offer crucial assistance for the reliability of the communications system. It also employs empirical study and a blend of primary and secondary methods to study its human resources. By listing the specific usage methods used by businesses and improving current human resource management issues to assist company in progressively addressing issues in the big era.

Enhancing the administration of the ability resource inside the company may also help the company to develop and create the potential resources in the company market leadership. The rapid expansion of data sources brought on by the automation of nearly all management structures has made a direct contribution to the advancement of technology for creating smart decisions (Hajimahmud et al., 2022).

Data mining, which is the act of examining data from many angles and condensing it into valuable information, is a new but powerful technique. The goal of data mining techniques is to extract information from the available data, which can then be used to process improvement. Due to the amount of data, data mining research has been drawn to the field of human resource management.

According to the study, HRM is a notable new area of data mining research that is predominantly technique- and technology-focused. However, particular domain criteria are necessary. Entrepreneurship and competitiveness are both constantly evolving as a result of reform. Additionally, the competitive paradigm has changed from being focused on material resources to being focused on people resources.

Consequently, human capital management (HCM) has emerged as a crucial connection for businesses. HCM is based on human resource planning. Through the unified gathering and administration of employee information, it may offer an overall assessment of the companies' internal employees' data. To address this issue, computer information technology has been developed. In recent years, data mining technology has advanced quickly. It is a technique for data analysis that might potentially retrieve information and knowledge.

The data mining technique has a wide range of applications because of its outstanding ability to process enormous amounts of data (Xu & Li, 2021). Data has grown in importance as an information system with the arrival of the big data revolution. It may enhance effectiveness and acquire insight into economic situations with the use of relevant data.

Technological resources may play a very high-value role in the development of company strategy and precise decision-making when big data technology is used to conduct research, evaluation, and mining of comprehensive information. Big data makes it possible to manage business human resource productivity creatively, which promotes the development of management style.

The majority of corporate departments are well aware of big data's uses because of its broad effects. The number of applications is constantly increasing in the age of big data and computer technology. Because of this, it can lessen the issues brought on by face-to-face contact and increase work productivity if it develops in human resource performance evaluation founded on these methodologies and the latest tools.

Connectivity may support the growth of positive company culture by becoming the standard medium of communication among workers, workers and managers, and consumers. As a result, they talked about the challenges involved in evaluating the effectiveness of HCM in the industry and agreed that doing so would significantly raise the bar for HCM organizations. This quantitative HCM examines data and investigates how data practices affect employee intentions and staff turnover.

To assess the successful applicability of big data extensive mining and analysis innovation in the overarching HRM, we proposed a binary crow search convolutional neural network (BCS-CNN) for the classification of big data in HRM. Big data features are extracted using linear discriminant analysis (LDA).

7.2 PROBLEM STATEMENT

A breakthrough in the growth of digital enterprises has been sparked by the advancement of science and technology, the arrival of the modern period, and the arrival of the big data era. HRM is a crucial division for online businesses (Snehal, Babasaheb & Khang, 2023a).

The absence of big data content approaches hinders the development of HRM studies and theories. Although academics have acknowledged the benefits of utilizing a big data strategy in HCM research, there is a lack of clear instructions on how to do so.

This chapter presents BCS-CNN for the classification of big data in HRM to lessen these drawbacks (Khang, Rani & Sivaraman, 2022).

7.3 PROPOSED WORK

With the implementation of a big data strategy in HCM data analysis by employing the suggested BCS-CNN for the classification of huge data, this chapter aims to further the development of big data in HCM. The dataset was assembled using the HCM database as a base (Babasaheb, Sphurti & Khang, 2023b).

The hybrid sine cosine genetic algorithm (HSCGA) is utilized in the HCM big data feature selection process. This section gives a systematic description, and Figure 7.2 shows the proposed method's flow.

7.3.1 Big Data Collection

For both business and everyday tasks, big data may offer greater analysis and decision-making support. Collected big data has the following characteristics: a large volume, diversity, value, speed, and accuracy of the employee, customer, and employer data (Rani et al., 2021).

Quantity refers to a sizable volume of data, which includes the quantity of information gathered, stored, and processed. Big data's first computation unit is at least 1,000 terabytes. It includes a range of sources and kinds of data, including those of geo-location, photos, recordings, sound scopes, and unorganized, partly organized, and organized data information (Bhambri et al., 2022). In the network era of digital economy, society effect subsequent 10,000 pieces of data information each day.

7.3.2 Data Preprocessing Using Z-Score Normalization

Data preprocessing is changing the numeric value of a specific dataset to streamline the content gathering and analysis. Normalization of the data reduces the difficulty of the method for processing it, since there is typically a very high disparity between the dataset's greatest and lowest values.

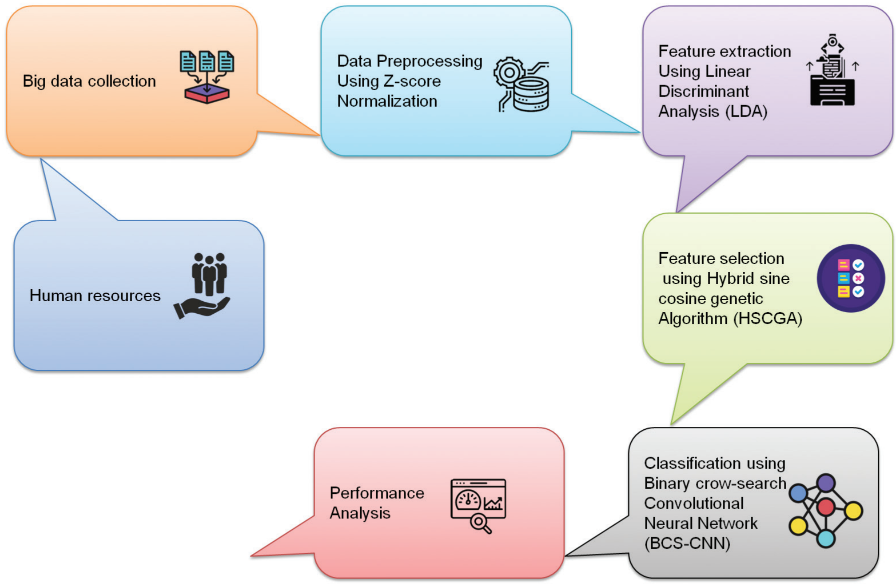

FIGURE 7.2 Flow of the proposed method.

The data can provide an appropriate advantage for the categorization of neural network-related algorithms, according to normalization. The retraining will go faster if the input values are normalized process. In this scenario if the feedback technique is utilized in node networks, making it a more effective neural network. Data attribute values are scaled so that they may be positioned in specific ranges (0 and 1) through the process of normalization (Khang et al., 2022).

Z-score normalization is based on the data's mean (overall average) and standard deviation (standard deviation), which may be determined by knowing the data's real lowest and highest values. Data scaling, which may transform the present range of data generally in the range [–1, 1] and [0, 1], is the foundation of the primary normalizing function. Equation 7.1 gives the normalizing formula.

$$q = \frac{(y - y_{min})(max - min)}{(y_{max} - y_{min}) + min} \tag{7.1}$$

The dataset's characteristics can be normalized using the standardization function, or z-score. It provides the ability to normalize a standard distribution by normalizing the attribute values of a collection. Equation (7.2), in which σ is the mean (an average value of an attribute in the dataset), and δ is the standard deviation of the mean, represents these values.

$$y'^{i} = \frac{y^{i} - \sigma^{i}}{\delta^{i}} \tag{7.2}$$

7.3.3 Feature Extraction Using Linear Discriminant Analysis

Massive computational complexity and a shortage of storage may be serious problems for big feature data analytics with large and high-dimensional data sets. This section starts by describing a feature extraction assignment for clarity. Many feature extraction approaches have been developed with LDA, which is one of the most fundamental and potent tools in the field.

We could utilize LDA for feature extraction (FE) issues when the dimension d of the data is bigger than the number N of the data due to the discontinuity of the within-class dispersion. In most cases, if d > N c, where c is the number of classes, a FE issue is referred to as a small sample size.

More significantly, it is possible to be certain that a vast amount of data contains a major component that is very evident. In other words, the big data conform to a Gaussian distribution and have a distinct main component, which is required by the LDA approach. This in turn implies that the LDA approach may be used to extract and classify big data.

The objective of LDA is to combine the initial predictors to produce a new variable. By maximizing the disparities between the preset groups relative to the new variable, this is achieved. The LDA method's two-class classification and multiclass classification each have a separate optimization function.

Assuming that the function must project into a less-feature space of integer values, the associated base vector is (w1, w2… wd), and the base vector composition matrix is W. It is the $n \times d$ matrix. Equation (7.3) states that T_a and T_v in Equation (7.4). The primary diagonal components of B are combined to form A. It may transform the $G(V)$ optimization procedure into an Equation.

$$T_a = \sum_{i=1}^{L} M_i(\mu_i - \mu)(\mu_i - \mu) \tag{7.3}$$

$$T_v = \sum_{i=1}^{L} \sum_{y=Y} Y(y - \mu_i)(y - \mu_i) \tag{7.4}$$

$$G(V) = \prod_{j=1}^{d} \frac{\omega_j^S T_a \omega_j}{\omega_j^S T_v \omega_j} \tag{7.5}$$

7.3.4 Feature Selection Using Hybrid Sine Cosine Genetic Algorithm (HSCGA)

Metaheuristic algorithms recently demonstrated remarkable performance in resolving practical issues. Binary search issues are referred to as feature selection issues. In the suggested approach, a consistent HSCGA is adjusted to deal with feature selection issues in the big data digital sector.

The search agent (r1) = 5, whereas the HSCGA starts with random places. Next, HSCGA functions following mathematical

$$u_1 = b - s\frac{a}{Smax} \tag{7.6}$$

$$g\theta = x * F = (1-x)\frac{\sum_j \theta j}{m} \tag{7.7}$$

When verifying and reaching the smallest number of selected feature choices simultaneously, the classifier over the validation data will function best, which will boost the fitness function of the HSCGA where n is the number of features in the dataset and the fitness function $g\theta$ returns a vector of size h with 0/1 components denoting unselected/selected features. F stands for the classifying failure rate, and Y is a constant used to manage how well the predictor performs given the chosen amount of features.

The variables employed share the same dataset's features count. The variables are constrained to the range [0, 1], where a variable value approaching 1 indicates that its associated features are candidates to be chosen for the classification.

In the genetic algorithm, a crossover is the top exploration operator. In light of existing solutions, it looks for potential solutions in the region. We used the genetic algorithm as a through using within the sine and cosine algorithm (SCA) as a hybrid feature selection method named HSCGA in the suggested hybrid plan to enhance the investigation tactic of the HSCGA and its efficiency during the search area for solving feature selection problems.

The suggested genetic algorithm enhances the present situation or response. As part of SCA, the crossover operator is used to create new offspring based on the best and candidate solutions, which are then improved by the mutation operator. Algorithm 7.1 describes the detailed procedure of HSCGA.

Algorithm 7.1: Hybrid Sine Cosine Genetic Algorithm

```
Create the SCA population from the beginning.
Equation 7.7 l=1 results in Fast Selection (FS) being the
optimal search agent.
        While (l<Max_iter+1)
               For i=1: N
                   Determine the optimal answer by calculating
u1, u2, u3, u4, and U1.
                   If (U1< 0.5)
                         Update the HSCGA by sine equation's
position.
                   Else
                         Update the HSCGA location using the
cosine equation.
                   End if
                   Improve using the suggested genetic
approach, the existing position.
```

```
                End For
                FS = the best search agent determined by
Equation (7.7);
        End While
        l = l + 1;
End For
```

7.3.5 Data Classification Using the Binary Crow Search Convolutional Neural Network (BCS-CNN)

A globally widely used network for tasks involving data mining, data processing, and classification is the convolutional neural network (CNN). In general, CNN is an artificial neural network that excels in picking up or detecting patterns and deciphering them. CNN is particularly helpful in picture analysis because it can discover patterns (Rana et al., 2021).

A CNN is a kind of ANN that differs from a traditional multilayer perception. Convolution layers, a hidden layer in CNN, are specifically capable of detecting ranges by selecting the number of nodes in one layer. Although CNN also uses additional non-convolutional layers, its core is made up of convolutional layers. The pooling layer function is to take in input and output, and transform input to the next layer for the convolution process (Khanh & Khang, 2021).

One of the most modern heuristic algorithms is the crow search algorithm (CSA). Crows are regarded as the brightest birds because they have the largest brains related to their physical size. Thus, we introduced the BCS-CNN for huge data categorization by fusing CNN with CSA.

The characteristics of BCS-CNN include flock shape, remembering the hiding sites, following one another during the heist, and guarding their belongings against an accidental theft. Crows are numerous in a dimensional setting the of architecture of BCS-CNN as shown in Figure 7.3.

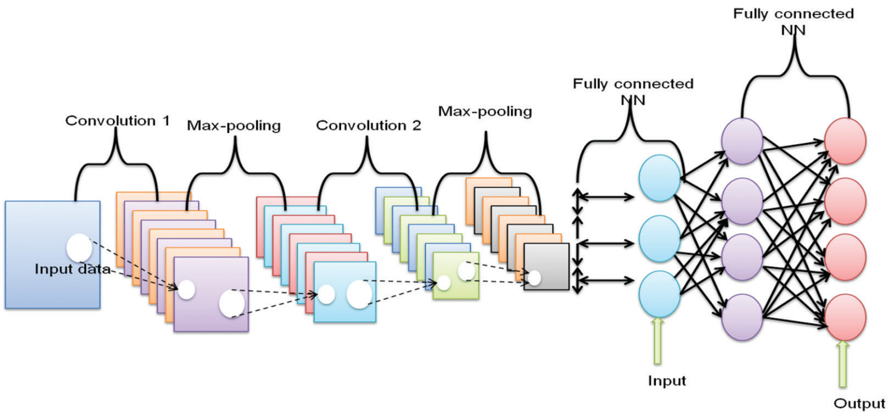

FIGURE 7.3 Architecture of BCS-CNN.

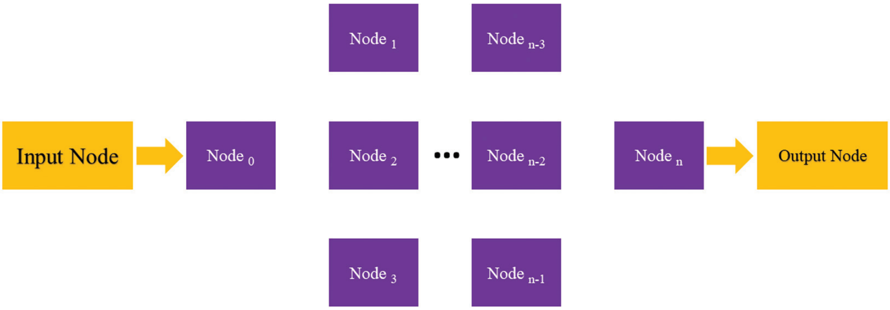

FIGURE 7.4 Classification nodes of BCS-CNN.

The maximal number of iterations is rept max, the number of data is M (size of data), the position of the data i on the recurrence classify area is vector-specified, and it displays the positioning of the data i's hiding spot in the rept. The data i has attained the best position, as indicated in Figure 7.4.

The following is the phasing procedure for implementing BCS-CNN into practice:

1. The customizable M, rept max, flen, and knowledge probability (KP) variables of the BCS-CNN are evaluated.
2. In a dimensional search field, M data points are distributed at random among the flock. Each set of data points to a workable answer, and it is the total number of choice factors.
3. Each piece of data must be added to the objective function of the choice variable value to calculate the standard of its location.
4. In the search space, data create a new spot as follows: Let's say that I wish to find a new place. To accomplish the new position of the data I will randomly choose data, and all of the field data were subject to this methodology.
5. Each data new area's fitness value is calculated and each data's storage has to be refreshed.

Algorithm 7.2: Binary Crow Search Convolutional Neural Network

```
Reset the position of all M data in the search field
Determine the position of the data
Change the memory of each data
while rept j rept max do
        for j=1:M do
                Choose one data to follow and define KP then
```

$$X^{i.rept\partial+1} = \mathrm{X}^{i.j.rept} + T_i \times flen^{i.rept} \times \left(m^{l.rept} - X^{i.rept}\right) \tag{7.8}$$

```
                Else
```

$$X^{i.rept\partial+1} \tag{7.9}$$

```
                        A is a random position of search space
Check feasibility of new location
Calculate the new position of the data
Update memory of data
```

7.4 PERFORMANCE ANALYTICS

Human capital management has received a great deal of attention across all businesses as an integral component of the top management paradigm. Due to the numerous data issues that human capital management encounters, it is critical to find insightful information and assist organizational decision-making. This chapter designs the use of big data comprehensive mining utilizing BCS-CNN as the analytical technique in the overall management of human resources.

This section evaluates the performance of the proposed BCS-CNN technique for the classification of big data in HCM. For the analysis of large data in human resource management, the suggested system can provide accuracy, precision, F1 score, and recall.

The existing methods are the ensemble classifier-decision tree (EC-DT) algorithm, back propagation neural (BPN) network, deep learning (DL), and simulated annealing algorithm (SAA).

7.4.1 Classification Accuracy

Classification accuracy is a measure to which the result of measurement corresponds with the appropriate value or a benchmark. Accuracy is the proportion of forecasts that the model successfully anticipated. The factor to consider while examining big data applications in HRM is accuracy.

Every business that needs to classify and minie the data must assess accuracy of the data. Especially comparisons between the approached process and other methodologies are made. Figure 7.5 displays the accuracy of the existing models and the suggested model accuracy. The suggested procedure works well for data classification in HRM.

The EC-DT achieved 50%, the BPN achieved 66%, the DL achieved 84%, the SAA achieved 75%, and the proposed method achieved 96%.

7.4.2 Precision

Precision is the level of intelligence that big data has incorporated. The ability of a big data model to analyze big data in human capital management is classified as precision. By dividing the entire number of genuine data values by the total number of true data plus erroneous data, precision is determined.

Out of all the examples, the model obtained the number of relevant instances. An evaluation of a model's precision in predicting positive labels is known as precision prediction. Figure 7.6 shows the existing model and the recommended model precision. The suggested procedure works well for data classification in HRM (Tailor, Pareek & Khang, 2022).

The EC-DT received 58%, the BPN received 78%, the DL received 88%, the SAA received 68%, and the suggested method received 95%, as shown in Figure 7.6.

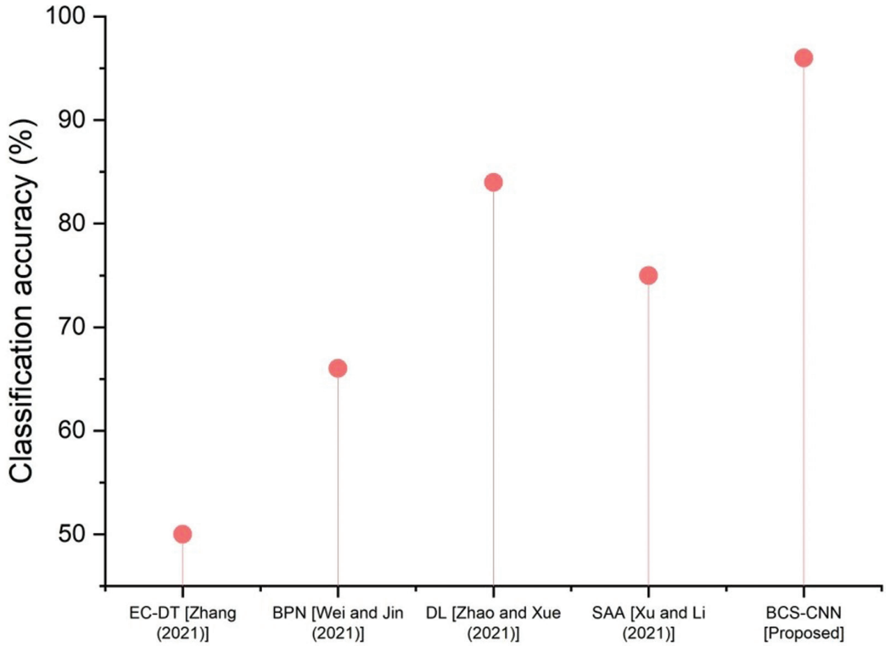

FIGURE 7.5 Classification accuracy of the existing and proposed model.

7.4.3 Recall

The recall is the ratio of the number of relevant data returned by classification to the total quantity of relevant data already in existence. The recall is calculated as a percentage of all positive data that were correctly classified as positive. How effectively the model can recognize positive data is shown by a recall.

The recall increases as more positive data are classified. A recall is more important in situations where missed data are more expensive than erroneous data. The existing system and the suggested model recall are depicted the classifying data in HRM in Figure 7.7.

7.4.4 F1-Score

By calculating their harmonic means, the F1-score combines a classification's accuracy and recall into a single statistic. Its main purpose is to evaluate the effectiveness of data classification. Using these criteria, the F1-score of a positive prediction is determined and divided by the entire quantity of positive prediction data.

The model with the highest recall value may also be the one that is the most responsive. The F1-score is the harmonic mean of recall and accuracy. In a single number, accuracy and recall are combined. Be aware that the F1-score takes

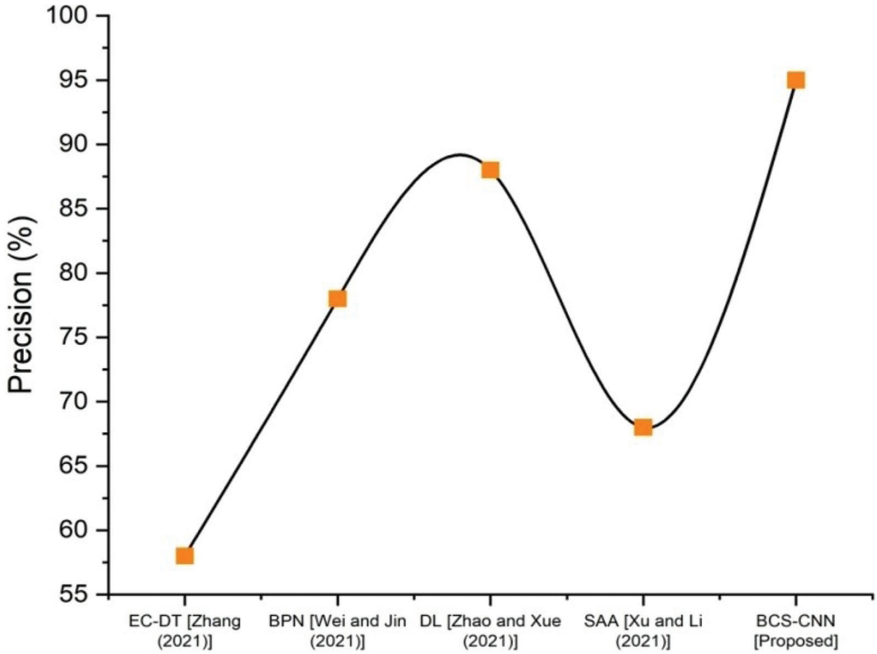

FIGURE 7.6 Precision of the existing and proposed model.

accuracy and recall into consideration, accounting for both false positive and false negative, as shown in Figure 7.8.

7.5 CONCLUSION

Human civilization has reached the era of the network economy as a result of constant advancement. The integration of big data with HCM will bring about significant shifts in thinking and even management in the era where big data is one of the new production aspects.

All HCM modules may now be classified as quantitative due to the idea of big data, which improves the effectiveness, precision, and specialization of HCM. HCM has a significant impact on how businesses grow.

To build a new management style from a global viewpoint, HRM uses the big data cognitive paradigm. As a result, this chapter suggested employing a BCS-CNN for huge data classification for the application of big data in HCM.

The findings demonstrate that the suggested BCS-CNN methodology performs well with the traditional EC-DT, BPN, DL, and SAA algorithm methods in terms of classification accuracy, precision, recall, and F1-score.

Future investigations into the subject under consideration can concentrate on enhancing the application of big data in HCM for effective management architectures. In the future, we may consider adding metaheuristic methods to increase

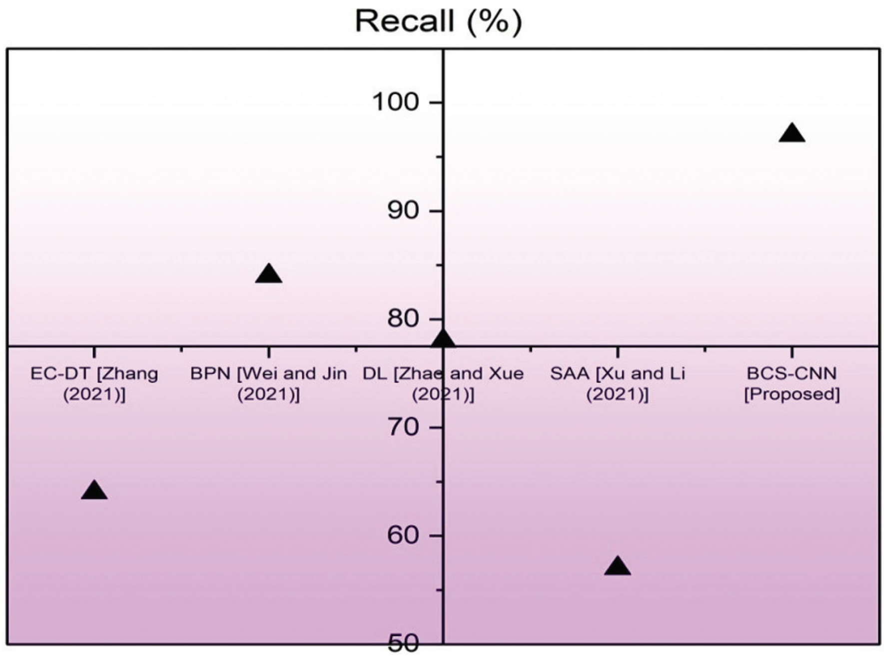

FIGURE 7.7 Recall the existing and proposed model.

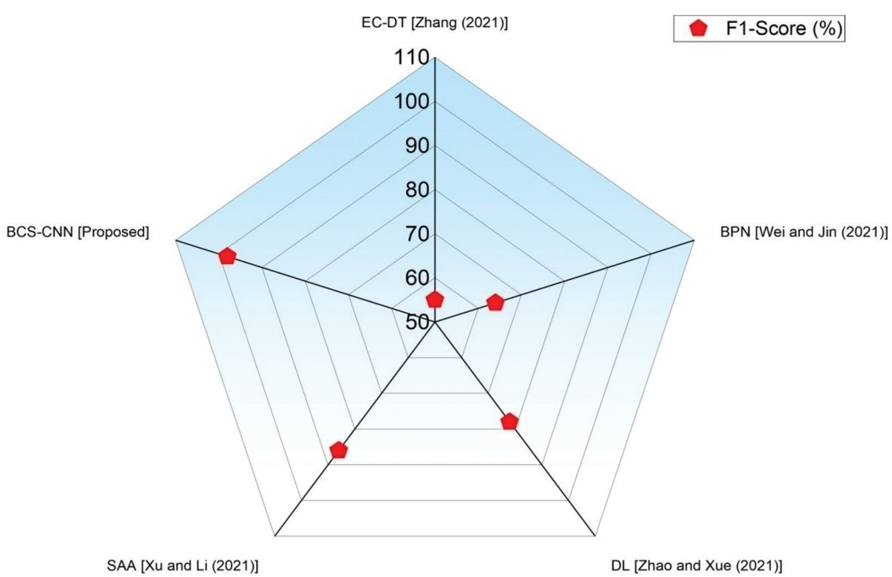

FIGURE 7.8 F1-score of the existing and proposed model.

performance metrics and the big data application in HCM domains by employing a range of methods (Khang et al., 2022a).

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8 Data-Centric Predictive Modeling of Turnover Rate and New Hire in Workforce Management System

Chandra Kumar Dixit, Parin Somani, Shashi Kant Gupta, and Anchal Pathak

CONTENTS

8.1 INTRODUCTION

Today's organizations place a premium on keeping a hold of their qualified human resources. (Dwesini, 2019) As a result, the organizational objectives are accomplished effectively and efficiently. In the fierce struggle to hire the most valued personnel in the market, maintaining human capital is crucial. Because they can transform other resources, like as cash, machinery, materials, and processes, into

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an output (a product or service), human resources are the key to gaining a competitive edge.

While other resources depreciated over time, human resources increased in value (Chin, 2018). These materials are unique, outstanding, and priceless. Nowadays, the vast majority of firms are aware that their human resources are the key factor in the success of their operations.

As a result, using humanistic management techniques to manage human resources has become essential for businesses to avoid absenteeism and turnover (Bamfo, Dogbe & Mingle, 2018). One of the most pressing problems in human resource management has been employee turnover.

Verma & Kesari (2020) has established strong roots in the organization's management procedures. The world has recently seen a rise in globalization, as the economies of many nations are linked to facilitate international commerce via connection and technology advancements.

This rise has also fueled increasing competition in domestic and international markets, which have further motivated businesses to hire and hold on to highly qualified individuals. For a competitive edge in the market, many businesses rely on their staff (Skelton, Nattress & Dwyer, 2019). As a result, they are closely tied to how well they manage and use their people resources as Figure 8.1 (Albalawi et al., 2019).

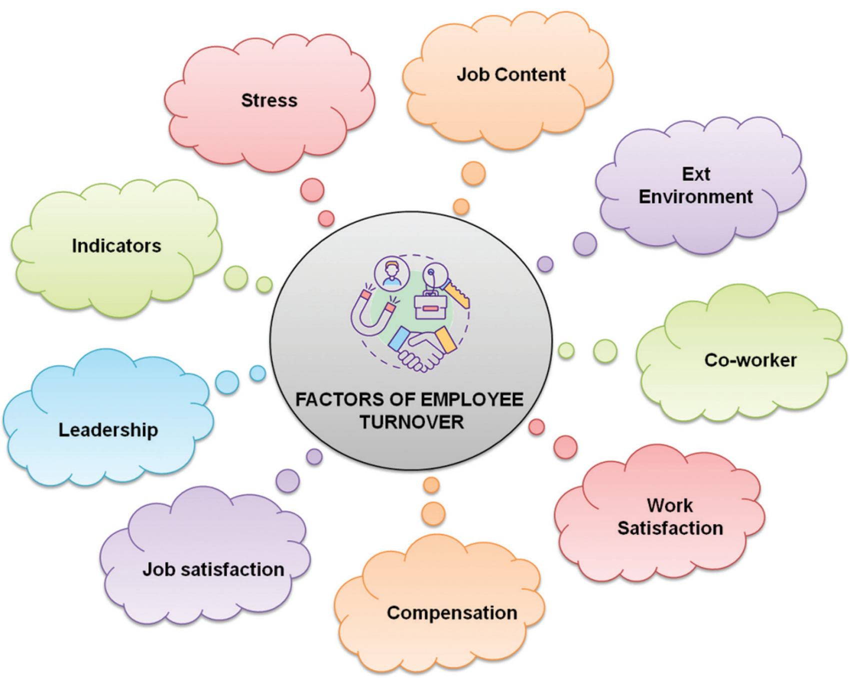

FIGURE 8.1 Factors of employee turnover.

8.2 PROBLEM STATEMENT

Some long-term care facilities have significant turnover rates among their personnel, which results in expensive business expenditures and diminished profitability and productivity. This problem is representative of a wider issue that affects businesses as a whole.

The unique commercial challenge is that the leaders of certain long-term care facilities have inadequate knowledge on the connection between pay, engagement, job satisfaction, motivation, and the work environment, which is one of the contributing factors to employee turnover (Babasaheb, Sphurti & Khang, 2023a).

The loss of staff due to attrition is a relatively unimportant factor for most businesses. It results in significant problems, such as a loss of financial resources, an increase in recruiting costs and time, discontent on the part of customers, and the need for further staff training.

Somehow, organizations may manage the loss of personnel due to attrition, even if those people do not have as much experience as employees who have been with the company for such a long period of time that their departure invariably results in some major financial losses.

8.3 PROPOSED WORK

We provide a new data-centric prediction model to solve these drawbacks using the evolutionary gradient boosted random forest algorithm (EGB-RFA). The company's human resource (HR) department's information was utilized to determine employee turnover (Khang et al., 2023).

As part of the preparation stage, we normalize the data to get rid of duplicates. The dataset assesses whether the employee is departing or remaining based on the characteristics. Now, we design a prediction model using 80% of the preprocessed data for training and 20% for testing. Principal component analysis (PCA) is used to extract the key traits (Snehal, Babasaheb & Khang, 2023).

The proposed framework is evaluated using the selected dataset with different feature settings and dataset sizes as shown in Figure 8.2.

8.3.1 Data Acquisition

In our experiment, we make use of a genuine dataset of workers from a China-based division of a communications corporation. The dataset has 2,000 workers and 25 attributes. Among these, 271 workers are designated as turnover, which indicates that they have checked the "attrition" boxes (Gupta et al., 2023).

The turnover rate, which is 13.5%, clearly indicates an issue with the balance of the data. The 25 qualities we chose as the most prevalent and widely known employee characteristics are included in Table 8.1 along with their respective ranges.

Table 8.1 shows the turnover rate, which is 13.5 percent, which clearly indicates an issue with the balance of the data.

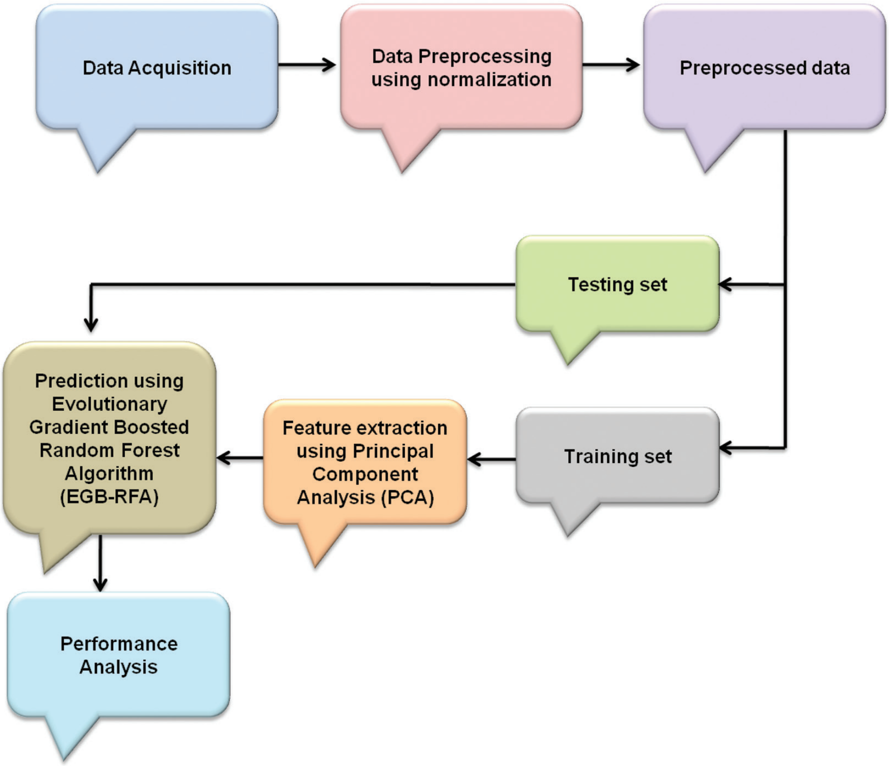

FIGURE 8.2 Proposed work of this research.

8.3.2 Data Preprocessing Using Normalization

Normalization of data is a preprocessing technique that entails rescaling or transforming the data so that each attribute contributes the same amount. The process of normalization involves the use of a scaling, mapping, or preprocessing technique to generate a new range from an existing one (Babasaheb, Sphurti & Khang, 2023b).

The advantages of prediction and forecasting might be substantial. Raw data may be transformed via a technique called normalization to ensure that each characteristic contributes the same amount. It deals with two key data worries that impede machine learning algorithms' learning process: The presence of dominant features and outliers.

The statistical measurements in the original (normalized) data have inspired a wide variety of methods for normalizing data within a certain range. The methods of min-max and z-score normalization are used in this investigation. Methods are categorized according to whether or not they use selected statistical properties of the raw data in the normalization process (Bhambri et al., 2022).

TABLE 8.1
Respective Ranges

#	Features	Value Range	Score
1	Employee Number	0–2000	0.0665
2	Age	22–60	0.0093
3	Gender	Men and women	0.0173
4	Job level	Low 4 to 13 High	0.0007
5	Job role	Staff, junior management, middle management, and senior management	0.0141
6	Education	1–20	0.2816
7	Monthly income	5,000–97,939	0.0015
8	Work life balance	Good, bad	0.0060
9	Job satisfaction	Low, medium, high, and very high	0.0137
10	Education field	Economics, management, technology, and other	0.2762
11	Overtime	Yes/no	0.0073
12	Number of companies worked	0–3	0.0289
13	Years in current role	0–15	0.0100
14	Winning count	0–5	0.0306
15	Percent salary increase	0–30	0.0031
16	Performance rating last year	Low, good, excellent, and outstanding	0.0257
17	Training time last year	0–5	0.0250
18	Year since last promotion	0–32	0.0218
19	Native place	Local, nonlocal	0.0007
20	Years in current management	0–12	0.0056
21	Physical condition	Healthy, unhealthy	0.0208
22	Employment nature	Regular worker, dispatched	0.0140
23	Total working years	0–37	0.0208
24	Department type	Sales, management, and technical	0.0140
25	Relationship satisfaction	Low, medium, high, and very high	0.0062

The min-max normalization technique provides a linear transformation of the initial data range. In other words, it's a method that keeps the connections between the real data intact. It's a basic method that may help you fit data into the ideal categories (Khang et al., 2023). As per this normalization method,

$$M' = \left(\frac{M - minvalue\ of\ K}{maxvalue\ of\ M - minvalue\ of\ M} \right) * (L - G) + G \tag{8.1}$$

Where M is a normalized set of min-max values. Limitation that has already been established is [G, L]: M represents the original data range, while G represents the mapped range. Parameter z-score normalization is a technique that creates standard values of data from unstructured data by using concepts like mean and standard